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Rapid response systems

Machine learning-based dispatch of drone-delivered defibrillators for out-of-hospital cardiac arrest



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Abstract

Background: Drone-delivered defibrillators have the potential to significantly reduce response time for out-of-hospital cardiac arrest (OHCA). However, optimal policies for the dispatch of such drones are not yet known. We sought to develop dispatch rules for a network of defibrillator-carrying drones.

Methods: We identified all suspected OHCAs in Peel Region, Ontario, Canada from Jan. 2015 to Dec. 2019. We developed drone dispatch rules based on the difference between a predicted ambulance response time to a calculated drone response time for each OHCA. Ambulance response times were predicted using linear regression and neural network models, while drone response times were calculated using drone specifications from recent pilot studies and the literature. We evaluated the dispatch rules based on response time performance and dispatch decisions, comparing them to two baseline policies of never dispatching and always dispatching drones.

Results: A total of 3573 suspected OHCAs were included in the study with median and mean historical ambulance response times of 5.8 and 6.2 min. All machine learning-based dispatch rules significantly reduced the median response time to 3.9 min and mean response time to 4.1–4.2 min (all $P < 0.001$) and were non-inferior to universally dispatching drones (all $P < 0.001$) while reducing the number of drone flights by up to 30%. Dispatch rules with more drone flights achieved higher sensitivity but lower specificity and accuracy.

Conclusion: Machine learning-based dispatch rules for drone-delivered defibrillators can achieve similar response time reductions as universal drone dispatch while substantially reducing the number of trips.

Keywords: Cardiac arrest, Drone, Automated external defibrillator

Introduction

Out-of-hospital cardiac arrests (OHCAs) occurring in residential and rural locations experience lower rates of survival compared

to those in other settings.^{1–5} For this population, rapid emergency medical service (EMS) response, initiation of bystander cardiopulmonary resuscitation, and access to automated external defibrillators (AEDs) continue to be significant challenges.^{3,6–12}

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Drones have increasingly been studied as a solution for rapid delivery of AEDs to OHCA. ^{13–17} Recent studies in rural and remote settings have focused on test flights to simulated OHCA by measuring the reduction in the time required for AED delivery ^{14,18} and demonstrating feasibility of drone-delivered AEDs. ^{18,19}

Despite these advances, many important questions remain regarding the implementation of drones within EMS systems. Dispatching is one such issue. EMS agencies must carefully consider the trade-offs that exist between the advantages of dispatching a drone to a given OHCA, which may reduce the response time of the OHCA, against the disadvantages. These include: (1) the cost of drone flights; (2) the possibility of adverse events or accidents; (3) the likelihood of the drone arriving after the ambulance; and (4) the unavailability of the drone for subsequent OHCA until its return from previous missions. To ensure their proper integration into EMS systems, dispatch policies for drones need to be developed and optimized, similar to what currently exists for ambulance dispatch.

In this paper, we develop and evaluate machine learning-based dispatch rules for drone-delivered defibrillators. We predict ambulance response times using supervised learning models and use these predictions to determine whether to dispatch drones based on a calculated drone flying time. We use OHCA data from Ontario, Canada, where drone feasibility studies are currently being carried out, ¹⁹ to evaluate the performance of these rules and characterize their trade-offs.

Methods

This study was approved by the research ethics boards at Sunnybrook Health Sciences Centre (Reference #: 286-2019) and the University of Toronto (Protocol #: 38278).

Study setting

The Regional Municipality of Peel (Peel Region) is a region in southern Ontario, Canada with a population of 1.38 million and land area 1247 km². It is further subdivided into the cities of Mississauga and Brampton, which consist largely of urban and suburban areas, and the town of Caledon, which is mostly rural.

Study population

We identified all suspected OHCA responded to by Peel Regional Paramedic Services between January 1, 2015–December 31, 2019. Suspected OHCA were defined as incidents assigned by EMS dispatch as cardiac arrest or unconscious. Incidents were excluded if the location of the suspected OHCA or the GPS-recorded starting location of the ambulance was not available, if the ambulance response time was not known, if the suspected OHCA incident occurred outside Peel Region, or if data was erroneous (e.g., response time and location data implied impossible vehicle speeds). Multiple ambulances were sometimes dispatched to the same incident. Ambulances that were dispatched more than 60 s after the first ambulance were excluded.

Drone dispatch rules

We developed a family of drone dispatch rules based on comparing a predicted ambulance response time plus a buffer time δ (in seconds)

to a calculated drone response time for a given suspected OHCA. For example, if $\delta = 60$, a drone is dispatched if it is predicted to arrive no more than 60 s after the ambulance. The δ parameter allows us to accommodate for delays in the actual ambulance response time (e.g., traffic congestion). We considered values of δ between 0 and 150 s in 30-s increments.

Drone response time calculation

We chose drone specifications based on those used in a current study in Peel Region, augmented with specifications from literature. ^{15,19} Ascent time, horizontal acceleration, cruising speed and descent velocity were set at 10 s, 19.6 m/s², 20 m/s and 2 m/s, respectively. The cruising altitude was set at 60 m, which is below the maximum altitude imposed by Transport Canada. Instead of attempting a full landing, the drone is designed to drop the AED in a protective casing from a height of 6 m; the time needed for the AED to reach the ground was set at 4.25 s.

The drone response time was defined as the time interval between drone take-off and the AED hitting the ground at the suspected OHCA location, using the straight-line distance between the two locations. This calculated drone response time was treated as the ground truth for evaluation, as there is no historical data on drone flights to the OHCA in our dataset.

Drone network design

We used a mathematical optimization model (Supplemental Methods) to determine optimal drone base locations such that the average response time to all suspected OHCA was minimized. Drone bases were chosen from EMS, fire, and police stations in Peel Region. We set the number of bases at three and assumed one drone was placed at each base. We conducted a sensitivity analysis by varying the number of drone bases between one and five. Model solutions were computed using Gurobi 9.0 optimization software (Gurobi Optimization LLC, Houston, TX).

Ambulance response time prediction

We developed linear regression and neural network (Supplemental Methods) models to predict ambulance response time, defined as the time interval between ambulance dispatch and arrival at the suspected OHCA location. The time of ambulance arrival on-scene was used instead of the time of first paramedic contact with the patient because we assumed that the time needed for a bystander to retrieve a drone-delivered AED was similar to a paramedic's on-foot travel time from the ambulance to the patient.

For linear regression, the independent variables were the road distance between the responding ambulance's location at the time of dispatch and suspected OHCA location, day of week categorized into three groups (Saturday, Sunday, and weekday), and time of day categorized into four groups (7:00AM–9:59AM, 10:00AM–3:59PM, 4:00PM–6:59PM, and 7:00PM–6:59AM). For the neural network, the independent variables were the road distance between the ambulance and suspected OHCA location, the day of week, the hour of day, the latitude and longitude of the suspected OHCA, and the latitude and longitude of the ambulance at time of dispatch. Road distances were computed using the OSMnx package ²⁰ within the Python 3.7.7 language (Python Software Foundation). If multiple ambulances were dispatched to a suspected OHCA, each ambulance's response time

was predicted individually and the minimum response time was used as the predicted response time to that suspected OHCA.

Dispatch rule evaluation

We evaluated the dispatch rules using five-fold cross validation: the models to predict ambulance response time were fitted using 80% of the data, and then predicted ambulance response times were generated for the out-of-sample 20%. This process was then repeated four more times each with a disjoint 20% out-of-sample dataset, resulting in out-of-sample predictions for the entire data set. Then, the dispatch rules were applied to the entire dataset using the predicted ambulance response times. We computed the distribution of out-of-sample *first response times*, defined as the minimum of the computed drone response time (if dispatched) and the actual ambulance response time based on historical data. We calculated the dispatch decision outcomes (Fig. 1), sensitivity, specificity, and accuracy for each dispatch rule by comparing each rule's dispatch decision to the decision that should have been made with foresight of the actual ambulance response time. In this context, sensitivity represents the percentage of incidents where a drone was dispatched, and it was faster than the ambulance. Specificity represents the percentage of incidents where a drone was not dispatched and would have arrived after the ambulance had it been dispatched. Accuracy represents the percentage of correct dispatch decisions.

We compared our dispatch rules to two baseline policies: (1) “never dispatch” (i.e., using historical ambulance response times), and (2) “always dispatch” (i.e., send the closest available drone to every suspected OHCA, regardless of whether it is predicted to be faster than the ambulance).

We tested whether the median first response time was less than the median historical ambulance response time using a right-tailed sign test, and whether the mean first response time was less than the mean historical ambulance response time using a right-tailed Wilcoxon signed-rank test. To determine whether our dispatch rules resulted in first response times that were not worse than the “always dispatch” policy, we conducted non-inferiority tests of the first response time between our drone dispatch rules and the “always dispatch” policy using a right-tailed sign test (for median) and a right-tailed Wilcoxon signed-rank test (for mean) with a non-inferiority margin of one second. A p-value less than 0.05 was considered significant for all tests.

We evaluated each dispatch rule in terms of drone availability at the time of each suspected OHCA. We assumed that a dispatched

Dispatch Decision	Outcome	
	Drone would have arrived before EMS	Drone would have arrived after EMS
	Send Drone	Don't Send Drone
Send Drone	True Positive	False Positive
Don't Send Drone	False Negative	True Negative

Fig. 1 – Outcomes for the drone dispatch decision. A false positive represents dispatching a drone that would have arrived after the ambulance. A false negative represents not dispatching a drone when it would have arrived before the ambulance.

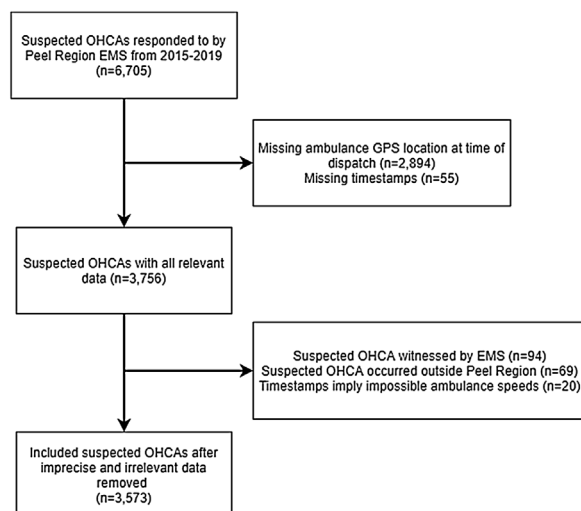


Fig. 2 – Inclusion criteria for suspected OHCA incidents. OHCA = out-of-hospital cardiac arrest.

drone would be occupied for the round-trip travel time between its base and suspected OHCA location, plus a 30-min reset time once the drone returned to its base to allow the drone to be charged and re-equipped with a new AED before taking off on another mission. If an ambulance arrived before the drone, we assumed the drone would turn around immediately and return to the base without reaching the suspected OHCA location. Each dispatch rule considers dispatching the closest drone to the suspected OHCA. If the closest drone is unavailable (e.g., is on another mission or is being reset), the next closest drone would be considered, and so on. If all drones were busy at the time of a suspected OHCA, then no drones would be dispatched to that incident. All evaluation metrics were computed based on the response time of the closest available drone. This dependency on drone availability means that for each dispatch rule, the dispatch decision outcomes are computed for a different set of optimal decisions, because each dispatch rule will have a different set of drones available for dispatch to each given suspected OHCA.

Because the decision to dispatch a drone to a suspected OHCA can affect subsequent drone availability, we calculated a metric called *harmful false positives*, defined as false positives that occupied a drone that, had it been available, would have been sent and improved response time to a subsequent suspected OHCA. This metric serves to quantify the detriment done to first response time by over-dispatching drones.

Table 1 – Incident characteristics for suspected OHCA victims during the study period. Number of missing data: age (n = 45), sex (n = 31), location (n = 15).

Characteristic	Included suspected OHCA victims (n = 3573)
Age in years, median (IQR)	69 (54–81)
Male	66 (52–79)
Female	74 (59–85)
Male sex, n (%)	2337 (66.0)
Public location, n (%)	503 (14.1)
Bystander witnessed, n (%)	1008 (28.2)
Bystander-initiated CPR, n (%)	914 (25.6)

Table 2 – First response time and evaluation metrics for each drone dispatch rule, assuming three drone bases are placed. The “never dispatch” rule corresponds to historical EMS response without drones.

Ambulance prediction model	Dispatch rule	Joint EMS and drone first response time (min)			Sensitivity, %	Specificity, %	Accuracy, %	Drone dispatches, n (%)	Suspected OHCA with improved response time, n (%)	Suspected OHCA where the closest drone was busy, n (%)	Harmful false positives, n
		Mean	Median	90th Percentile							
–	Never dispatch	6.2	5.8	9.5							
Linear regression	$\delta = 0$ s	4.2	3.9	6.7	93.8	76.0	87.8	2508 (70.2)	2218 (62.1)	211 (1.7)	3
	$\delta = 30$ s	4.1	3.9	6.6	97.0	67.4	86.9	2684 (75.1)	2289 (64.1)	216 (1.8)	6
	$\delta = 60$ s	4.1	3.9	6.5	98.7	56.6	84.4	2855 (79.9)	2328 (65.2)	219 (1.9)	7
	$\delta = 90$ s	4.1	3.9	6.5	99.4	47.0	81.6	2986 (83.6)	2342 (65.6)	221 (1.9)	11
	$\delta = 120$ s	4.1	3.9	6.5	99.6	38.5	78.7	3093 (86.6)	2343 (65.6)	225 (2)	14
Neural network	$\delta = 150$ s	4.1	3.9	6.5	99.7	30.8	76.2	3190 (89.3)	2346 (65.7)	225 (2)	15
	$\delta = 0$ s	4.2	3.9	6.7	93.5	77.1	87.9	2487 (69.6)	2210 (61.9)	210 (1.6)	4
	$\delta = 30$ s	4.1	3.9	6.6	96.4	68.6	87.0	2657 (74.4)	2276 (63.7)	216 (1.8)	7
	$\delta = 60$ s	4.1	3.9	6.5	97.9	59.6	84.9	2801 (78.4)	2311 (64.7)	217 (1.8)	8
	$\delta = 90$ s	4.1	3.9	6.5	99.2	51.0	82.8	2933 (82.1)	2337 (65.4)	220 (1.9)	10
–	$\delta = 120$ s	4.1	3.9	6.5	99.5	41.6	79.8	3055 (85.5)	2344 (65.6)	223 (2)	13
	$\delta = 150$ s	4.1	3.9	6.5	99.6	34.7	77.4	3141 (87.9)	2344 (65.6)	225 (2)	14
	Always dispatch	4.1	3.9	6.5	100.0	0.0	65.7	3573 (100)	2349 (65.7)	233 (2.3)	20

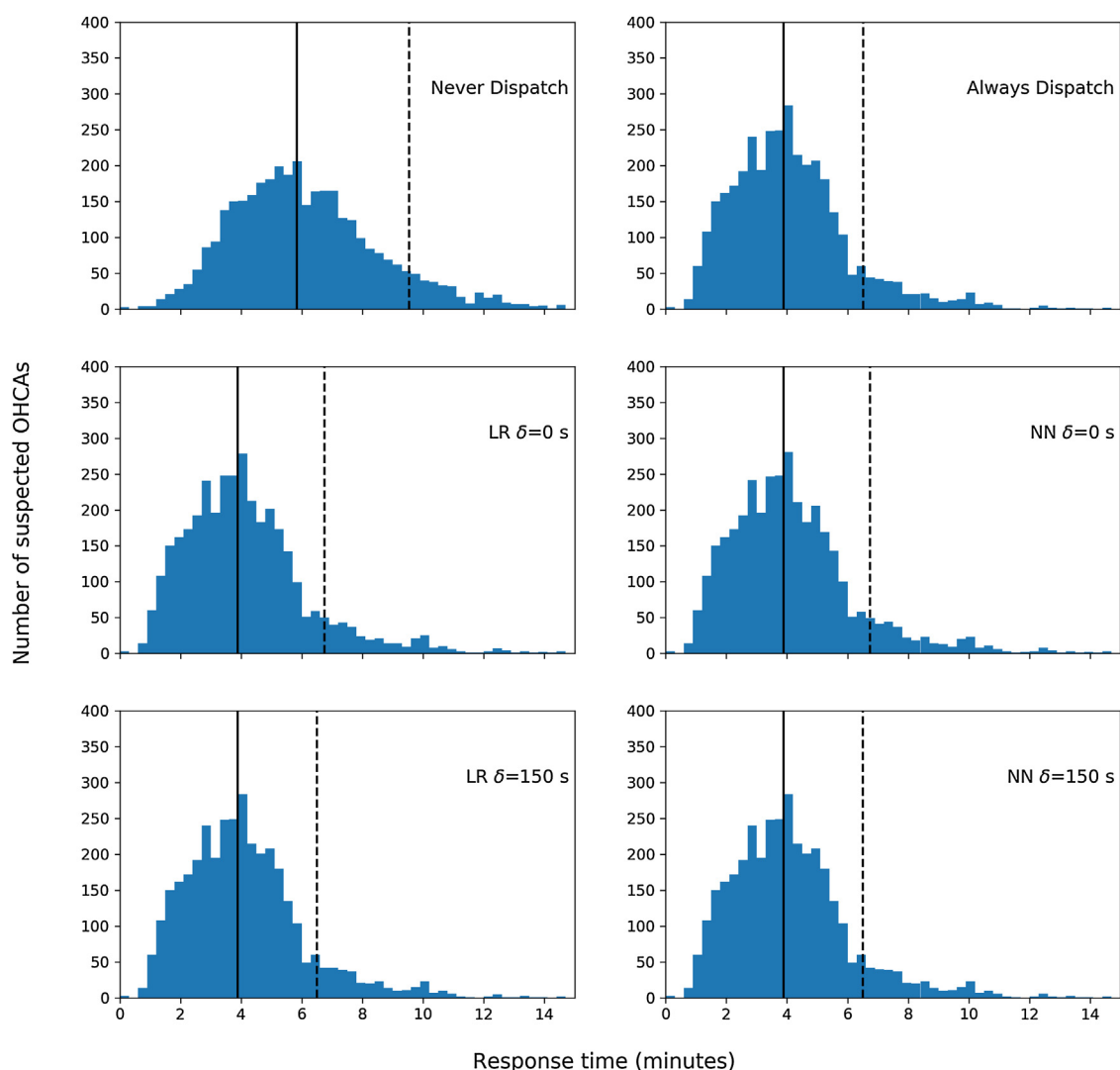


Fig. 3 – The response time distributions for the “never dispatch” (i.e., historical EMS response), “always dispatch,” and linear regression (LR) and neural network (NN) dispatch rules, with buffer (δ) times of 0 and 150 s and three drone bases. The response time distributions for all other values of δ are shown in Supplemental Fig. 1. The solid line represents the median of the distribution and the dashed line represents the 90th percentile.

Results

We identified 6705 suspected OHCA's that occurred during the study period. Of those, 3573 suspected OHCA's met the inclusion criteria (Fig. 2). Table 1 summarizes the patient characteristics of our included study population.

The mean, median, and 90th percentile historical ambulance response times (i.e., “never dispatch”) were 6.2, 5.8, and 9.5 min, respectively (Table 2). For all dispatch rules, the mean, median and 90th percentile first response times were 4.1–4.2, 3.9, and 6.5–6.7 min, representing improvements of approximately 2–3 min for each metric. The mean and median first response times were significantly faster than historical ambulance response times (all $P < 0.001$) and were non-inferior to the “always dispatch” rule (all $P < 0.001$). Fig. 3 shows the full response time distribution for the two baseline policies and several of the machine learning-based dispatch

rules. Not only are the median and 90th percentile response times of our dispatch rules similar to the “always dispatch” rule, but their response time distributions are also nearly identical.

Table 2 also compares dispatch rules in terms of sensitivity, specificity, accuracy, and several other metrics. For example, the dispatch rule utilizing linear regression with $\delta = 0$ dispatched drones to 70.2% of all suspected OHCA's and achieved 93.8% sensitivity, 76.0% specificity, and 87.8% accuracy. As δ increased, the dispatch rate and sensitivity increased, but specificity and decision accuracy decreased. Large values of δ resulted in many additional drone dispatches, marginally improving sensitivity but substantially degrading specificity and accuracy. When drones were dispatched more frequently, the frequency of the closest drone being busy and the number of harmful false positives increased, but remained low, while the percentage of suspected OHCA's where a drone improved response time initially increased and then plateaued. Our sensitivity analysis on the number of drone bases produced similar results (Supplemental Tables 1 and 2).

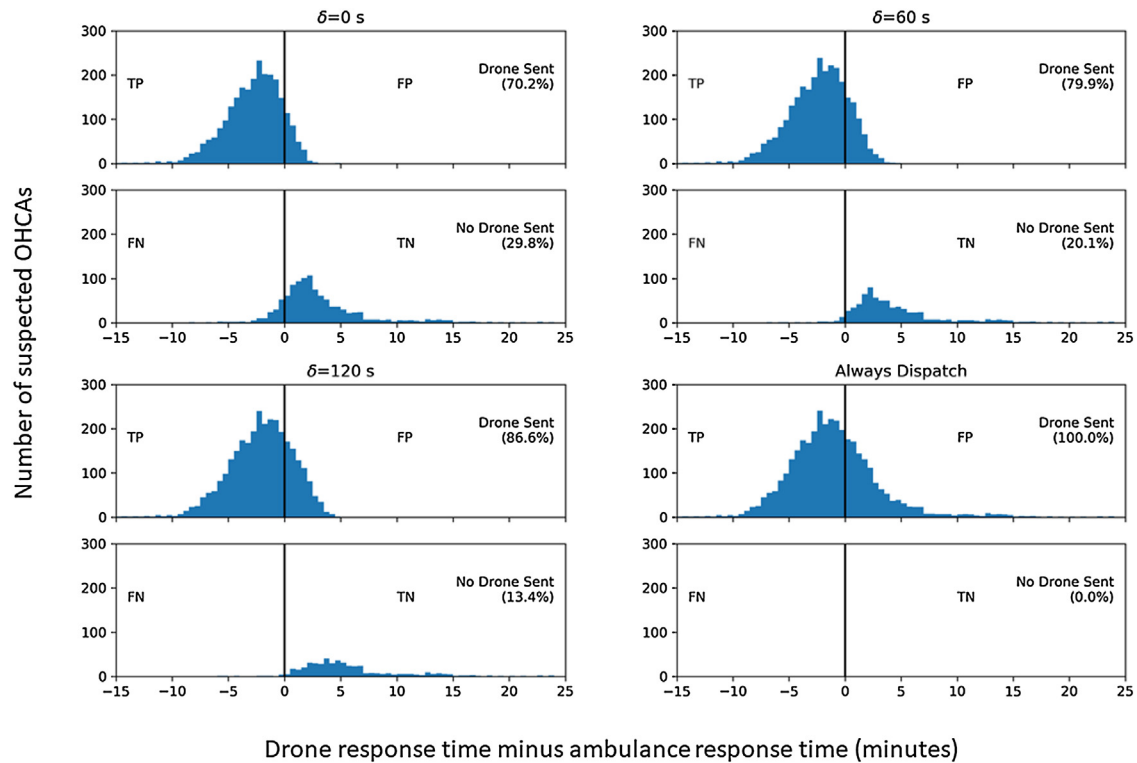


Fig. 4 – Drone response time minus actual ambulance response time stratified by dispatch decision for the linear regression dispatch rules with $\delta = 0, 60$, and 120 s, and the “always dispatch” rule, assuming three drones were placed. As the value of δ increases, the number of False Negatives decreases, and the number of False Positives increases.

Fig. 4 visualizes the time interval between the arrival of the ambulance and the arrival of the drone stratified by the drone dispatch decision for several of the linear regression-based dispatch rules. Similar figures for the remaining rules are shown in Supplemental Fig. 2. Positive values denote the drone arriving at the location of the suspected OHCA after the ambulance, and negative values denote the drone arriving prior to the ambulance. As δ increased, the distributions shifted to the right, resulting in more false positives but fewer false negatives.

Discussion

This is the first study of dispatch rules for AED-carrying drones to suspected OHCA's. We found that dispatch rules that selectively dispatch a drone to a suspected OHCA, based on the prediction that it reaches the incident before an ambulance, can achieve a similar distribution of first response times compared to a policy that dispatches a drone to all OHCA's. The intuition behind this result is that as drones are dispatched more frequently, there is both an increase in the likelihood of drones being unavailable and a reduction in the number of incidents where the first response time can be improved. Our findings serve as an important foundation for EMS agencies to determine the dispatch of drones such that resources are used both effectively and efficiently.

Consistent with prior literature,^{13–15,18,19} we find that drone-delivered AEDs may significantly reduce the time needed for AED arrival on scene. In our study, we found a 2-min reduction in both the mean and median first response time. Based on the tenet in existing literature that each one-minute delay in the time to

defibrillation resulted in a 10% decrease in the odds of survival,²¹ we would expect an increase of around 20% in the odds of survival, assuming that improvement in the first response time results in the same improvement in the time to defibrillation. Previous drone research for OHCA's focused on the optimal location of bases and the allocation of drones to bases, assuming that drones would be sent to every OHCA. Our novel finding in this paper is that if dispatch decisions are made with the aid of machine learning models, response time improvements can be maintained with up to 30% fewer dispatches. Reducing the number of dispatches is likely to improve the cost-effectiveness of a drone network and reduce the risk of adverse events such as equipment failures and collisions. Further studies are needed to determine the full cost-effectiveness of drone dispatch rules when integrated within a specific EMS system.

The dispatch rules we developed included a parameter δ , which allows a decision maker to modulate conservatism in drone dispatch. A larger value of δ represents a willingness to dispatch drones more frequently, leading to more cases where a drone improves the response time but also more cases where the drone was unnecessary (Table 2). This parameter can be adjusted by EMS agencies to achieve a desired balance between sensitivity and specificity of dispatch decisions, weighing the benefit of improved response time for some patients against the cost of unnecessary dispatches to others. Our results suggest a δ value between 0 and 90 s provides reasonable balance between these conflicting objectives (Supplemental Fig. 3), but the generalizability of this finding to other jurisdictions requires additional study.

Fig. 4 shows that most of the false negatives (i.e., an incorrect decision to not send a drone) and false positives (i.e., an incorrect decision to send a drone) for our dispatch rules have drone and

ambulance response times that are similar, suggesting that a large portion of incorrect dispatch decisions were incorrect by only a small margin. Thus, the high accuracy of our dispatch rules (Table 2) is bolstered by the fact that even when the dispatch rule made an incorrect decision, the average error was relatively small.

Both families of models considered (linear regression and neural network) produced comparable response time predictions, which resulted in comparable dispatch decisions and first response time distributions (Fig. 3). From an implementation perspective, linear regression-based dispatch rules are likely preferable because they are simpler to model and interpret.

For a drone-delivered AED initiative to be successful, several barriers must be overcome. First, the concept requires community buy-in, not just through acceptance of drones flying through the area but also by increasing the number of lay responders who are willing to use the delivered AED; this may be achievable through targeted education and social marketing campaigns.^{22–24} Second, the logistics of operating the drone network, such as operating and maintaining the drones, ensuring that drone-delivered AEDs are returned, and having trained staff who can keep the drone network running, must be developed. Third, regulatory constraints must be satisfied: drones would be required to be flown beyond line-of-sight, currently permitted in some regions but not others; and issues regarding flight clearance and priority between drones and other aircraft would need to be resolved. Careful coordination across governmental agencies will be required such that AEDs can be delivered safely and efficiently.

Our study has several limitations. First, we did not consider factors that may affect the drone response time such as weather, reliability of the drone, routing, or obstacles in the flight path. Second, we did not include response times from police, fire, or subsequently dispatched ambulances who may also attend to the suspected OHCA. Including these response times may reduce the benefit that drones can provide. Third, due to incomplete and erroneous data (Fig. 2), our analysis only included 53% of the total suspected OHCAs during the study period. This suggests that our estimates for drone unavailability are underestimated for all dispatch rules. As a result, a more conservative dispatch rule may be warranted.

Conclusion

We developed machine learning-based decision rules to efficiently determine whether to dispatch AED-carrying drones to suspected OHCAs, which can achieve similar response time reductions as universal drone dispatch but with a substantial reduction in the number of drone flights. The dispatch rules contain a tolerance parameter that EMS agencies can adjust to achieve a desired balance between dispatch decision sensitivity and specificity.

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Conflict of interest statement

Dr Cheskes has received speaking honorarium for educational events on cardiopulmonary resuscitation quality from Zoll Medical and

Stryker Corporation. Dr Cheskes has received grant funding from Zoll Medical for AED on the Fly, Community Responder Program for Peel Region, and Monitoring Ventilation During OHCA research studies. Dr Cheskes sits on the Advisory Board of Drone Delivery Canada. The other authors have no conflicts of interest to report.

CRedit authorship contribution statement

Jamal Chu: Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft. **K.H. Benjamin Leung:** Methodology, Software, Validation, Formal analysis, Investigation, Writing - original draft. **Paul Snobelen:** Resources, Data curation. **Gordon Nevils:** Resources, Data curation. **Ian R. Drennan:** Resources, Writing - review & editing, Project administration. **Sheldon Cheskes:** Conceptualization, Writing - review & editing, Supervision, Funding acquisition. **Timothy C.Y. Chan:** Conceptualization, Methodology, Writing - review & editing, Supervision.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.resuscitation.2021.02.028>.

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